assignment 11 - Faraday;s law of induction

This is a preview of the draft version of the quiz

Started: Mar 23 at 6:48pm

Quiz Instructions

Based on lectures. See slides. Remember to convert.

Question 1 3.2 pts

A step-up transformer steps up voltage by ten times. Neglecting slight losses, if 100 W of power go into the primary coil, the power coming from the secondary coil is

- 100 W.
- 1 W.
- none of the above
- 10 W.
- 1000 W.

Question 2 3.2 pts

The voltage across the input terminals of a transformer is 120 V. The primary has 50 loops and the secondary has 25 loops. The voltage the transformer delivers is

- none of the above
- 240 V.
- 120 V.
- 60 V.
- 30 V.

Question 3 3.2 pts

A certain transformer doubles input voltage. If the primary coil draws 10 A of current, then the current in the secondary coil is

- [ ] none of the above
- [ ] 10 A.
- [ ] 25 A.
- [ ] 5 A.
- [ ] 2 A.

Question 4 3.2 pts

Disconnect a small-voltage battery from a coil of many loops of wire and a large voltage may be produced by

- [ ] the resistance of the battery to a change in polarity.
- [ ] the electric field between the battery terminals.
- [ ] latent energy in the battery.
- [ ] the sudden collapse in the magnetic field.
- [ ] electrons already in the wire.

Question 5 3.2 pts

Underlying the concept of self-induction is

- [ ] Maxwell's law.
- [ ] Faraday's law.
- [ ] all of these about equally
- [ ] Ohm's law.

Question 6 3.2 pts

Electromagnetism unites

- [ ] electric and gravitational fields.

electricity and magnetism.

all of the above

none of the above

physics and biophysics.

Question 7 3.2 pts

A transformer actually transforms

all of the above

non-safe forms of energy to safe forms of energy.

magnetic field lines.

generators into motors.

voltage.

Question 8 3.2 pts

A step-up transformer steps up voltage by ten times. If voltage input is 120 volts, voltage output is

1200 V.

12000 V.

none of the above

60 V.

120 V.

Question 9 3.2 pts

The output power of an ideal transformer is

greater than the input power.

equal to the input power.

smaller than the input power.

may be any of the above

Question 10 3.2 pts

The rate at which energy is transferred is

power.

transformation.

none of the above

electromagnetic induction.

Question 11 3.2 pts

If the voltage produced by a generator alternates, it does so because

the changing magnetic field that produces it alternates.

unlike a battery, it produces alternating current.

of alterations in the mechanical energy input.

none of the above

all of the above

Question 12 3.2 pts

The current produced by a common generator is

dc.

ac.

neither of these

Question 13 3.2 pts

A device that transforms mechanical energy into electrical energy is a

generator.

- magnet.
- transformer.
- motor.
- none of the above

Question 14 3.2 pts

A device that transforms electrical energy to mechanical energy is a

- none of the above
- magnet.
- motor.
- transformer.
- generator.

Question 15 3.2 pts

The metal detectors that people walk through at airports operate via

- civil laws.
- Coulomb's law.
- Faraday's law.
- Newton's laws.
- Ohm's law.

Question 16 3.2 pts

If you drop a bar magnet in a vertical copper pipe it will fall slowly because

- it induces a magnetic field in the pipe that resists motion of the magnet.
- the copper is a good conductor of both electricity and magnetism.

of air resistance.

of electron repulsion.

Question 17 3.2 pts

When a change occurs in the magnetic field in a closed loop of wire

none of the above

a current is created in the loop of wire.

a voltage is induced in the wire.

electromagnetic induction occurs.

all of the above

Question 18 3.2 pts

The frequency of induced voltage in a wire coil depends on

how frequently a magnet dips in and out of the coil.

the frequency of current producing it.

the number of loops in the coil.

Question 19 3.2 pts

When a magnet is moved to and fro in a wire coil, voltage is induced. If the coil has twice as many loops, the voltage induced is

none of the above

the same.

twice.

four times as much.

half.

Question 20 3.2 pts

Magnetic field strength inside a current-carrying coil will be greater if the coil encloses a

- vacuum.
- glass rod.
- wooden rod.
- rod of iron.
- none of the above

Question 21 3.2 pts

Electromagnetic induction occurs in a coil when there is a change in

- magnetic field intensity in the coil.
- none of the above
- electromagnetic polarity.
- electric field intensity in the coil.

Question 22 3.2 pts

When a bar magnet is thrust into a coil of copper wire, the coil tends to

- repel the magnet as it enters.
- both of these
- attract the magnet as it enters.
- neither of these

Question 23 3.2 pts

If you thrust a magnet into a closed loop of wire, the loop will

- rotate.
- have a current in it.
- both of these

neither of these

Question 24 3.2 pts

A flat circular loop of radius 0.10 m is rotating in a uniform magnetic field of 0.20 T. Find the magnetic flux through the loop when the plane of the loop and the magnetic field vector are parallel.
hint: there is a flux when the field lines are going through the area only

5.5×10^{-3} T · m^2

6.3×10^{-3} T · m^2

0 T · m^2

3.1×10^{-3} T · m^2

Question 25 3.2 pts

A flat circular loop of radius 0.10 m is rotating in a uniform magnetic field of 0.20 T. Find the magnetic flux through the loop when the plane of the loop and the magnetic field vector are perpendicular.
hint: use the equation of the flux.

3.1×10^{-3} T · m^2

0 T · m^2

5.5×10^{-3} T · m^2

6.3×10^{-3} T · m^2

Question 26 3.2 pts

A flat circular loop of radius 0.10 m is rotating in a uniform magnetic field of 0.20 T. Find the magnetic flux through the loop when the plane of the loop and the magnetic field vector are at an angle of 30°.
hint: use the equation for the flux. (cos(60) because you take the angle swept by the area from its position perpendicular to B).

3.1×10^{-3} T · m^2

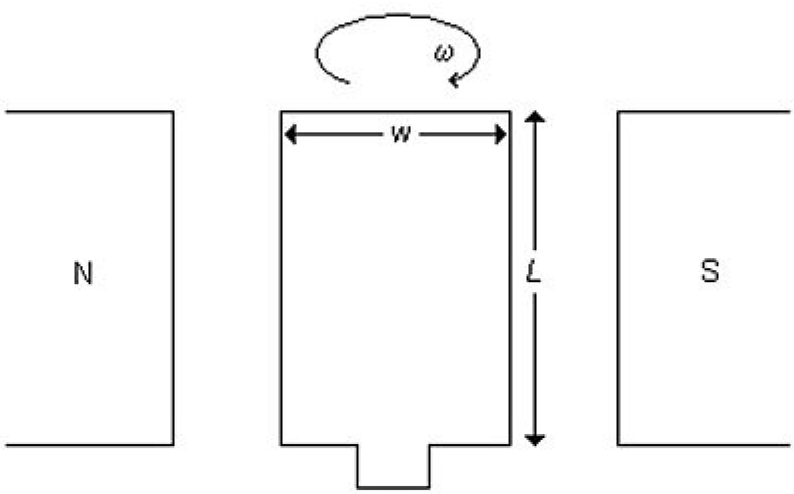
0 T · m^2

6.3×10^{-3} T · m^2

5.5×10^{-3} T · m^2

Question 27 3.2 pts

A rectangular loop of wire that can rotate about an axis through its center is placed between the poles of a magnet in a magnetic field with a strength of 0.40 T, as shown in the figure. The length of the loop L is 0.16 m and its width w is 0.040 m. What is the magnetic flux through the loop when the plane of the loop is perpendicular to the magnetic field?



0 T · m^2

2.6×10^{3} T · m^2

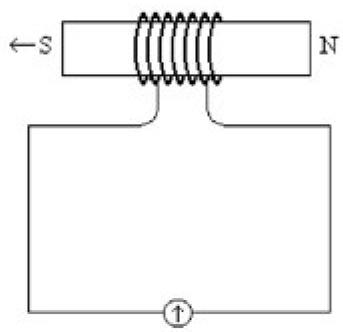
2.6×10^{-3} T · m^2

13×10^{-3} T · m^2

0.80 T · m^2

Question 28 3.2 pts

A bar magnet is pushed through a coil of wire of cross-sectional area 0.020 m^2 as shown in the figure. The coil has seven turns, and the rate of change of the strength of the magnetic field in it due to the motion of the bar magnet is 0.040 T/s. What is the magnitude of the induced emf in that coil of wire?



- 5.6×10^{-3} V
- 5.6×10^{-5} V
- 5.6×10^{-4} V
- 5.6×10^{-2} V
- 5.6×10^{-1} V

Question 29 3.2 pts

A flux of 4.0×10^{-5} T · m^2 is maintained through a coil of area 7.5 cm^2 for 0.50 s. What emf is induced in this coil during this time by this flux?

- 4.0×10^{-5} V
- No emf is induced in this coil.
- 2.0×10^{-5} V
- 3.0×10^{-5} V
- 8.0×10^{-5} V

Question 30 3.2 pts

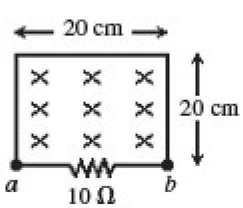
A closed flat loop conductor with radius 2.0 m is located in a changing uniform magnetic field. If the emf induced in the loop is 2.0 V, what is the rate at which the magnetic field strength is changing if the magnetic field is oriented perpendicular to the plane in which the loop lies?

- 0.16 T/s
- 0.080 T/s
- 2.0 T/s
- 1.0 T/s

Question 31 3.2 pts

As shown in the figure, a wire and a 10-Ω resistor are used to form a circuit in the shape of a square with dimensions 20 cm by 20 cm. A uniform but non-steady magnetic field is directed into the plane of the circuit. The magnitude of the magnetic field is steadily decreased from 2.70 T to 0.90 T in a time interval of 96 ms. What is the induced current in the circuit, and what is its direction through the resistor?

hint:Remember Ohm's law. V = R I. and also. the current indiced will flow in such a way that it will create a magentic field that opposes the first one. So the induced magnetic field is toward you.



- 75 mA, from *b* to *a*
- 45 mA, from *b* to *a*
- 45 mA, from *a* to *b*
- 110 mA, from *a* to *b*
- 75 mA, from *a* to *b*

Question 32 3.2 pts

A round flat conducting loop is placed perpendicular to a uniform 0.50 T magnetic field. If the area of the loop increases at a rate of 3.0×10^{-3} m^2/s, what is the induced emf in the loop? hint: use the equation. Now it is the area that is changing.

- 1.5 mV
- 0 mV
- 4.3 mV
- 5.5 mV
- 1.7 mV

Not saved Submit Quiz